

Lead Score Case Study Summary

Problem Statement

X Education specializes in selling online courses to industry professionals and is seeking assistance in identifying the most promising leads—those with the highest probability of converting into paying customers.

The company requires a model that assigns a lead score to each lead, with the aim of prioritizing customers. Customers with higher lead scores are expected to have a greater likelihood of conversion, while those with lower lead scores are anticipated to have a lower chance of conversion.

The CEO, specifically, has provided an approximate target for the lead conversion rate, aiming for approximately 80%.

Solution

Reading and Understanding Data

Read and analyze the data.

Data Cleaning

Dropped the variables with a significant percentage of missing values. This process also involved imputing missing values, when necessary, using median values for numerical variables and generating new categorical variables. Outliers were detected and subsequently handled.

Data Analysis

Subsequently, initiated the Exploratory Data Analysis (EDA) to gain insights into the data's distribution and structure. In the data analysis, there are variables which has no specific inferences or conclusions can be drawn due to limited variability. So, these variables were dropped.

Preparation of Data

Proceeded by converting Yes/No values to 1/0 for better data analysis and generating dummy variables for the categorical features.

Splitting the Data into Training and Testing Sets

This step involved splitting the dataset into training and testing subsets, with a 70-30% proportion.

Rescaling Features

Applied Standard Scaling to standardize the original numerical attributes. Following that, employed the stats model to build initial model, providing a comprehensive statistical overview of all model parameters.

Performing Feature Selection with Recursive Feature Elimination (RFE)

- Utilized Recursive Feature Elimination (RFE) to identify the top 24 important features.
- Using the statistics generated, iteratively examined the p-values to select the most significant variables, while discarding the insignificant ones.
- In the end, we identified the 12 most significant variables, and the Variance Inflation Factors (VIF) for these variables were found to be acceptable.
- Generated a dataframe that included converted probability values. Initially, assumed that a probability value exceeding 0.5 would be considered as 1, while those below 0.5 would be labeled as 0.
- Following the aforementioned assumption, calculated the Confusion Matrix and determined the overall Accuracy of the model.
- Additionally computed the 'Sensitivity' and 'Specificity' metrics to assess the model's reliability.

Plotting the ROC Curve

Next, constructed an ROC curve for the features, which displayed a commendable performance with an area under the curve (AUC) of 89%. This further bolstered the model's robustness.

Finding Optimal Cutoff Point

We subsequently generated probability graphs for 'Accuracy,' 'Sensitivity,' and 'Specificity' across various probability values. The point where these graphs intersected was identified as the optimal probability cutoff point, which turned out to be 0.37.

With the new cutoff value, we observed that the model correctly predicted nearly 80% of the values.

The new values for accuracy, sensitivity, and specificity are as follows: accuracy=81.09%, sensitivity=81.73%, specificity=80.69%.

Metrics- Precision and Recall

The Precision and Recall metrics values on the train dataset were found to be 79.54% and 70.60%, respectively.

Making predictions on the Test Set

After applying the model to the test dataset, the accuracy value was found to be 80.50%, with Sensitivity at 80.49% and Specificity at 80.51%.